KiCad Tool Training 2/2

Layout Editor & **Gerber Viewer**

KiCad Version 8.0

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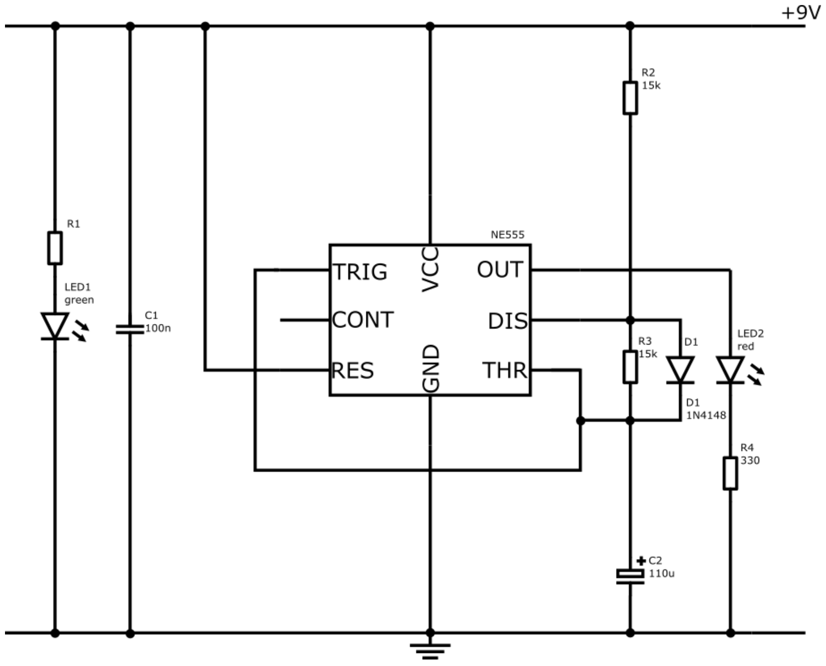
University of Applied Science Darmstadt

2024



Layout Constraints

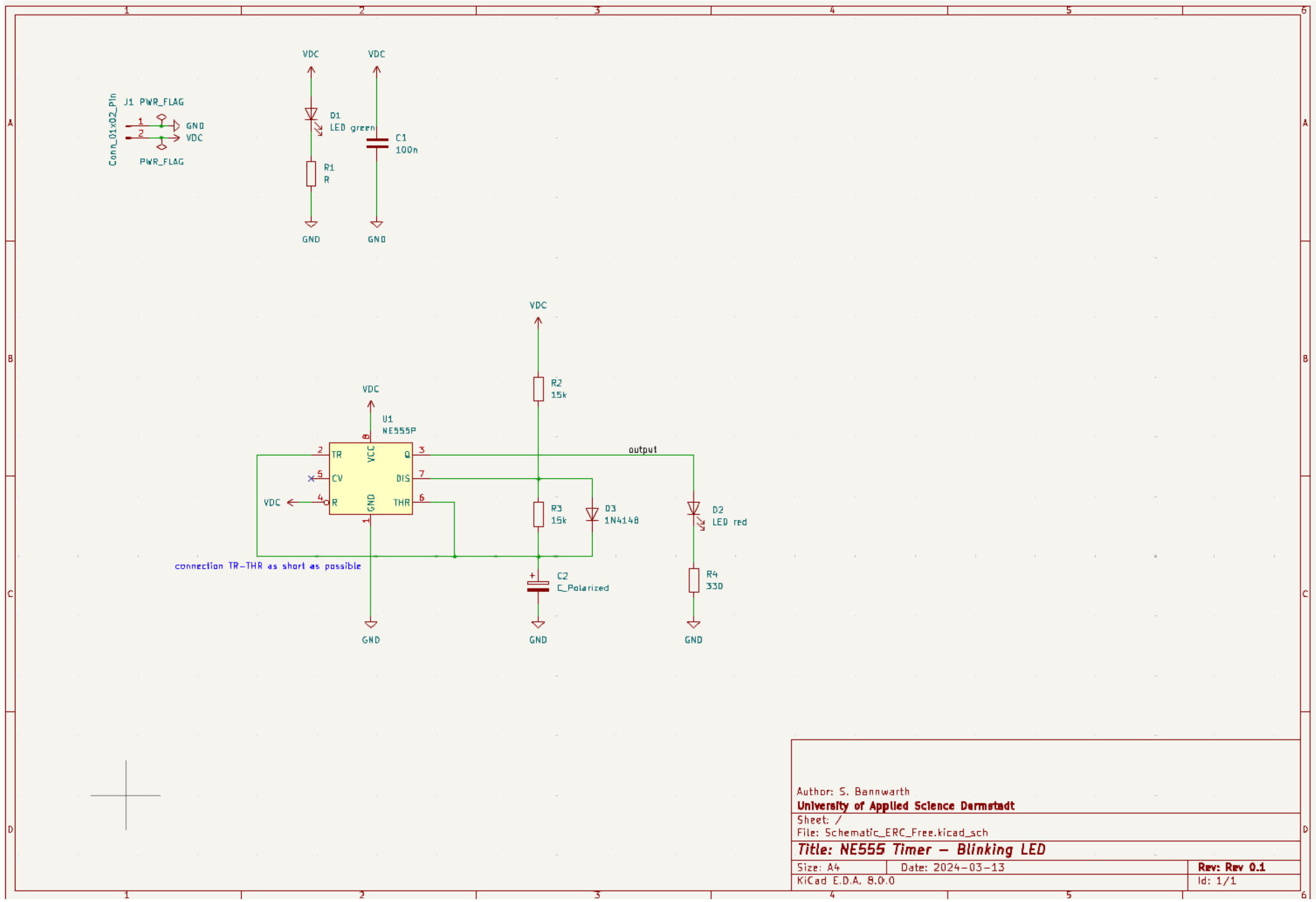
Board Size:	52mm x 25mm
4x Mouting holes:	M3
Layer:	2 copper layers
Top Layer:	signal routing
Bottom Layer:	GND – plane

Agenda	Working Steps
Layout Constraints	1. Update your Schematic 2. Clarify PCB Constraints 1. Your own 2. Manufacturers
Mechanical Aspects	3. Board Dimension 4. Mounting holes 5. Import netlist
Routing	6. Place Components 7. Resolve Crossings 8. Routing 9. Fill Free Areas
Prepare for Production	10. Add Project Information on the top Silk Layer 11. Perform the Design Rule Check 12. Export to Gerber 13. Check Gerber Files

Agenda

- Layout Task and its Working Steps
- Updating your Schematic
- Layout Constraints
- Mechanical Aspects
- Routing
- Prepare for Production
- Exercise
- Engineering Best Practice

Update your Schematic



Checks

1. Anything changed?
2. What's about testing?
3. What's about assembling?

Notes

1. Layout Constraints

Agenda

- Layout Task and its Working Steps
- Updating your Schematic
- Layout Constraints
- Mechanical Aspects
- Routing
- Prepare for Production
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- Engineering Best Practice

Project Layout Constraints

Board Size:	52mm x 25mm
4x Mouting holes:	M3
Layer:	2 copper layers
Board Thickness:	1.6mm
Top Layer:	signal routing
Bottom Layer:	GND – plane

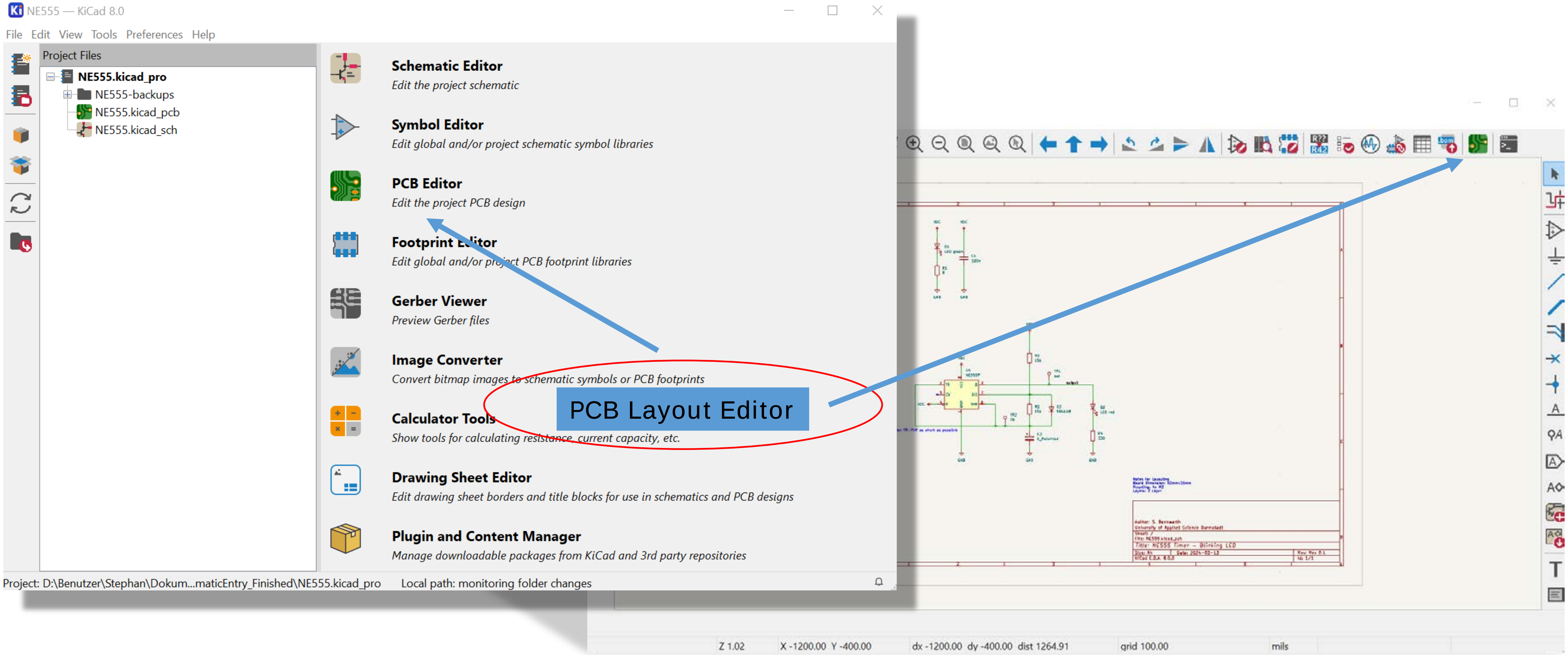
PCB Manufacturer

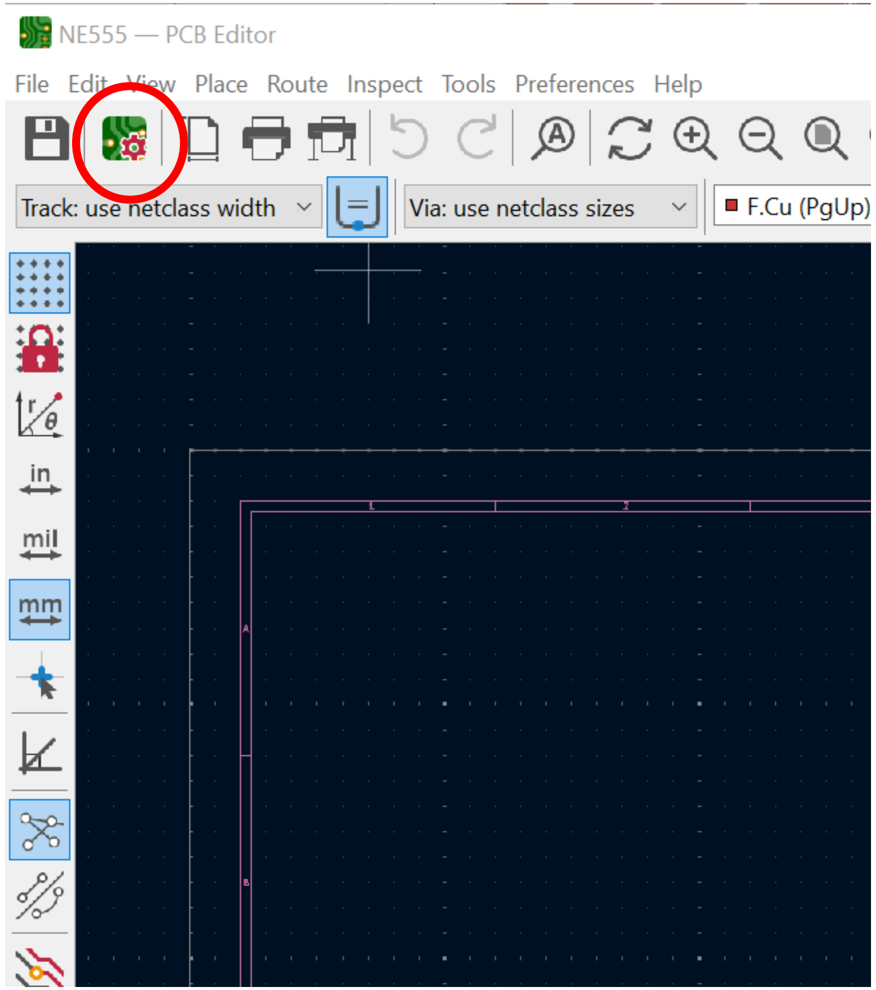
https://www.elekonta.de/

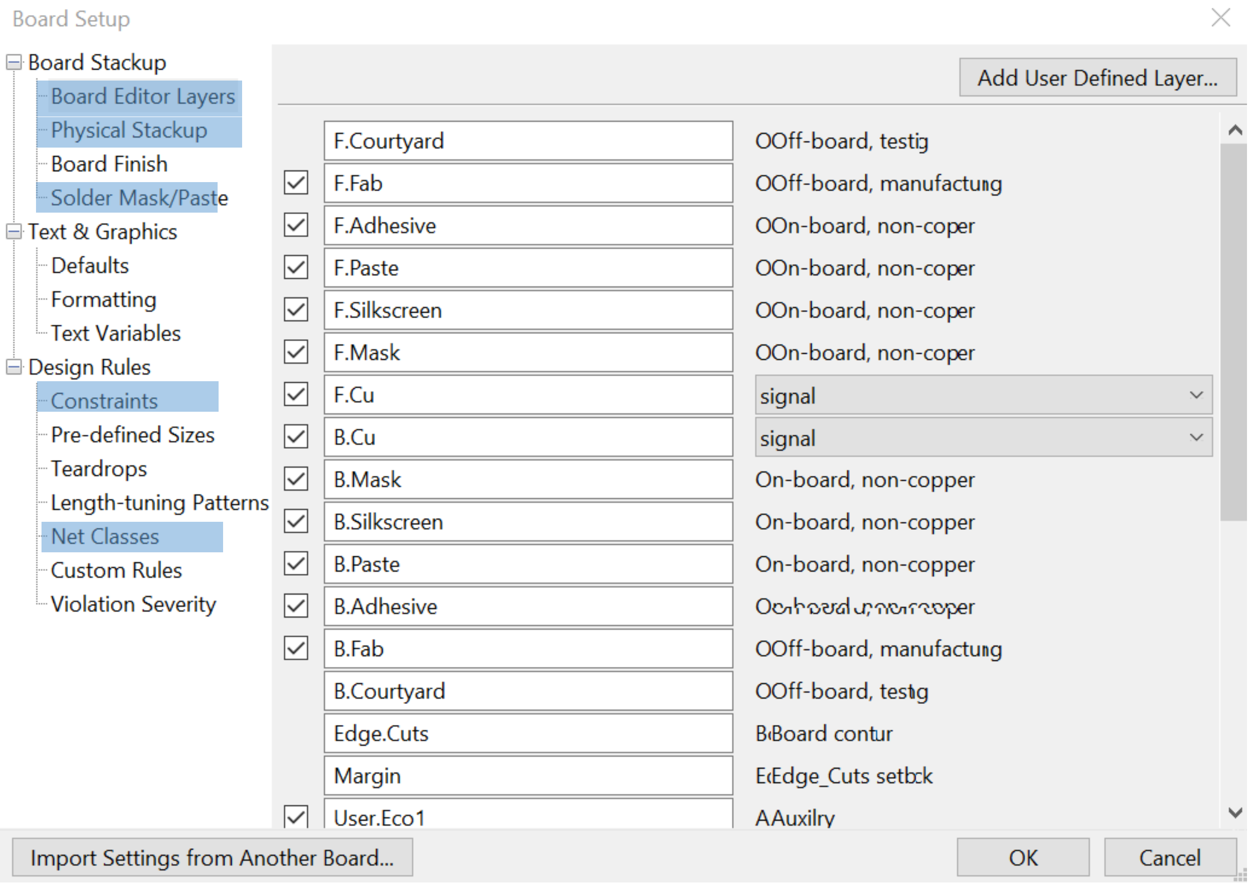
Base Material Manufacturer

https://www.isola-group.com/

Layout Editor




NE555 — PCB Editor
File Edit View Place Route Inspect Tools Preferences Help
Track: use netclass width
Via: use netclass sizes
F.Cu (PgUp)
in
mil
mm

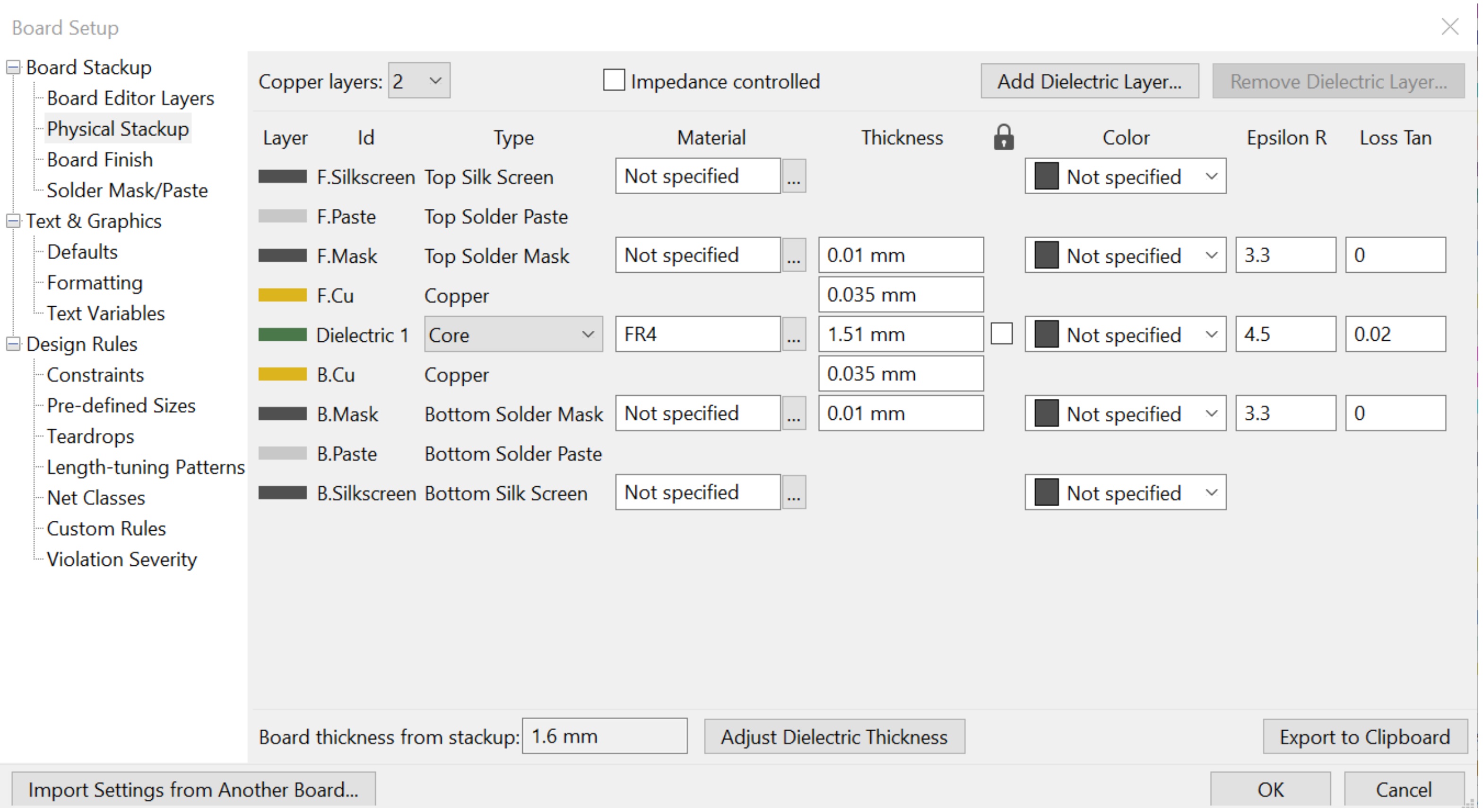

Board Setup
Board Stackup
Board Editor Layers
Physical Stackup
Board Finish
Solder Mask/Paste
Text & Graphics
Defaults
Formatting
Text Variables
Design Rules
Constraints
Pre-defined Sizes
Teardrops
Length-tuning Patterns
Net Classes
Custom Rules
Violation Severity
Add User Defined Layer...
F.Courtyard
F.Fab
F.Adhesive
F.Paste
F.Silkscreen
F.Mask
F.Cu
B.Cu
B.Mask
B.Silkscreen
B.Paste
B.Adhesive
B.Fab
B.Courtyard
Edge.Cuts
Margin
User.Eco1
signal
signal
On-board, non-copper
Import Settings from Another Board...
OK
Cancel

KiCad

1. Manufacturer Constraints

→ Check Manufacturers' Website
→ Contact for Specification

Assume Board-Thickness: 1.6 mm



2. Your Project Constraints

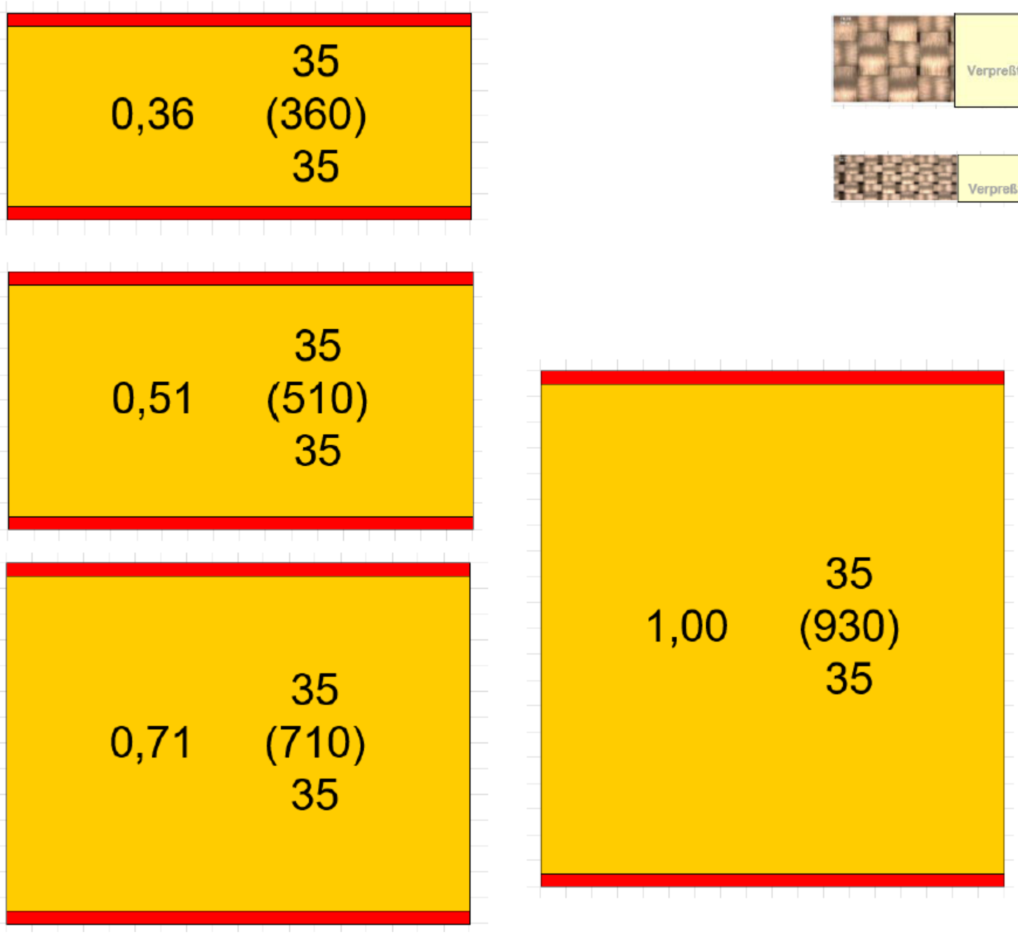
Board Size:	52mm x 25mm
4x Mouting holes:	M3
Layer:	2 copper layers
Top Layer:	signal routing
Bottom Layer:	GND – plane

Isola IS400 Materialauswahl

Isola IS400 material selection

Basismaterial

Laminate / Core



Prepreg

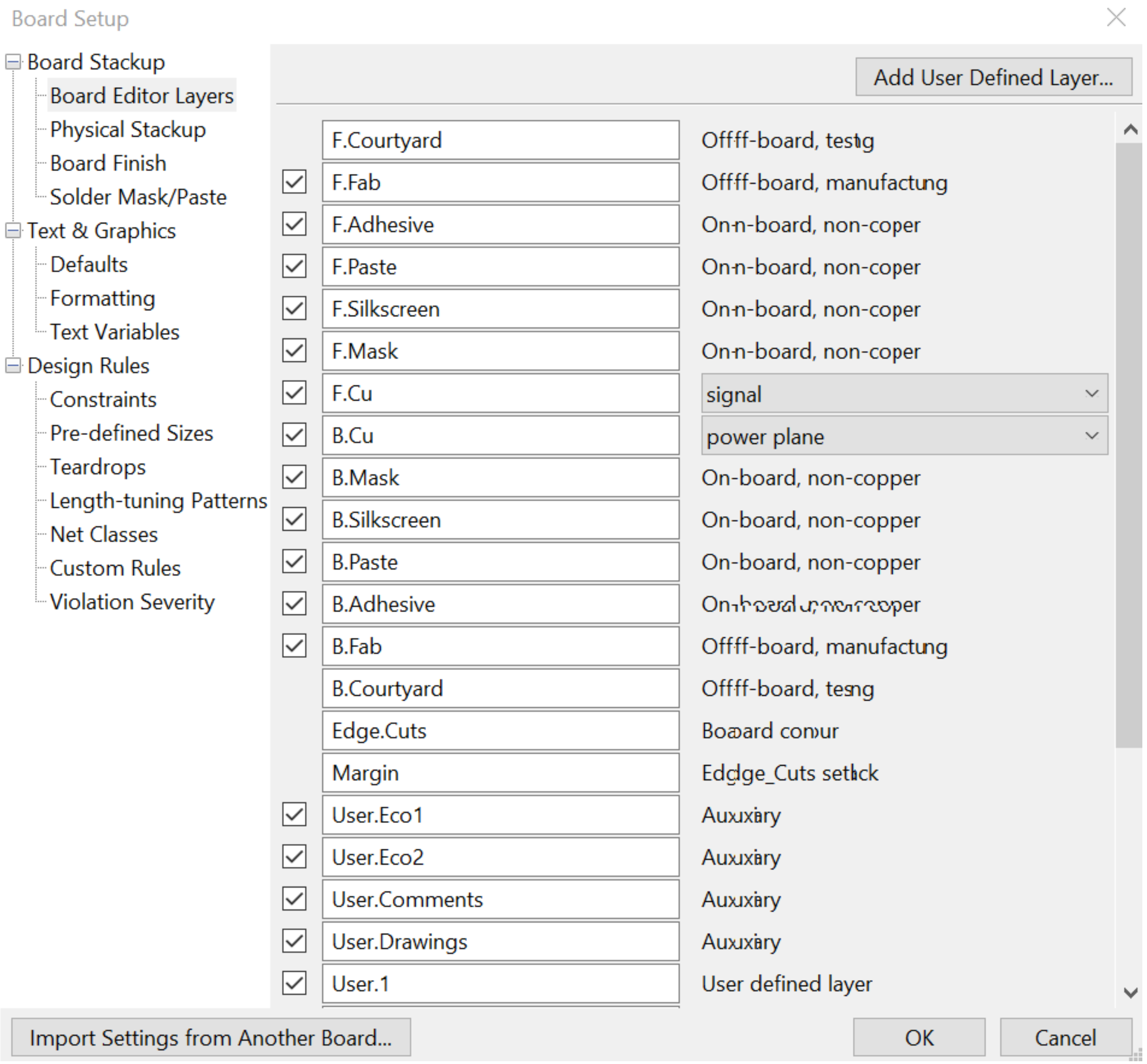
Laminate / Core

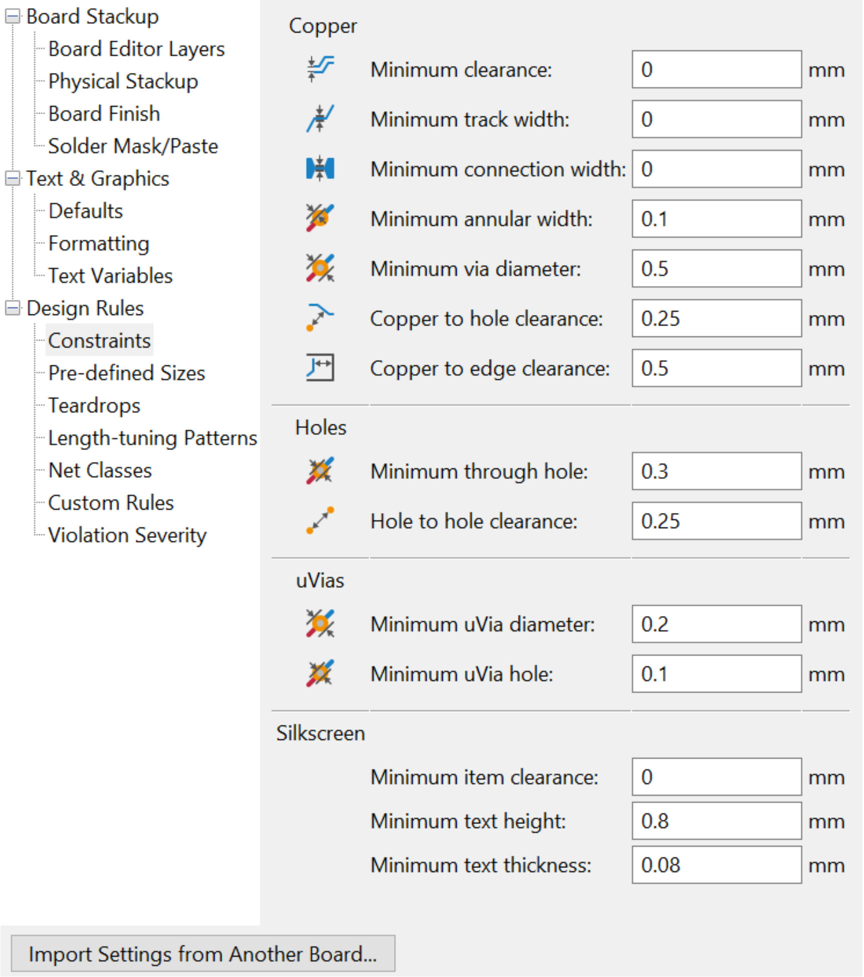
ELEKONTA MAREK

Cu-Folie / galv.Cu / Lötstoplack

Cu-Foil / electroplated Cu / Solder resist

Board Setup Menu → Board Editor Layers





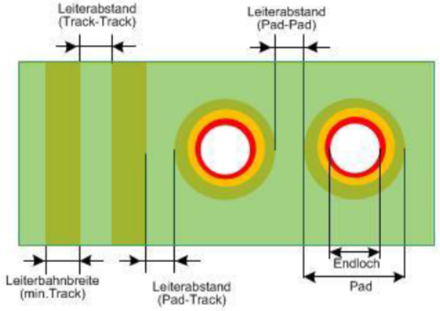
Manufacturer Constraints

Outer layer (Outer), galv. Inner layer

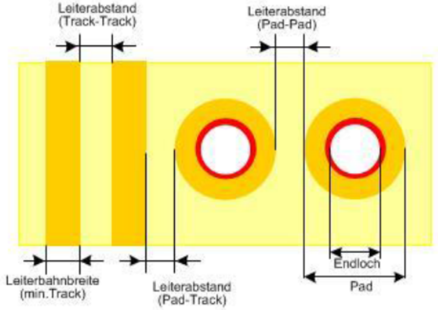
Basic copper	Pad – Pad* Pad – Track* Track – Track* (Standard)	Min.Track* (Standard)
12µm	75µm	75µm
12µm (with MicroFill)	100µm	100µm
18µm	100µm	100µm
35µm	125µm	125µm
...		

* Smaller values are possible.

Außenlage /Outer layers

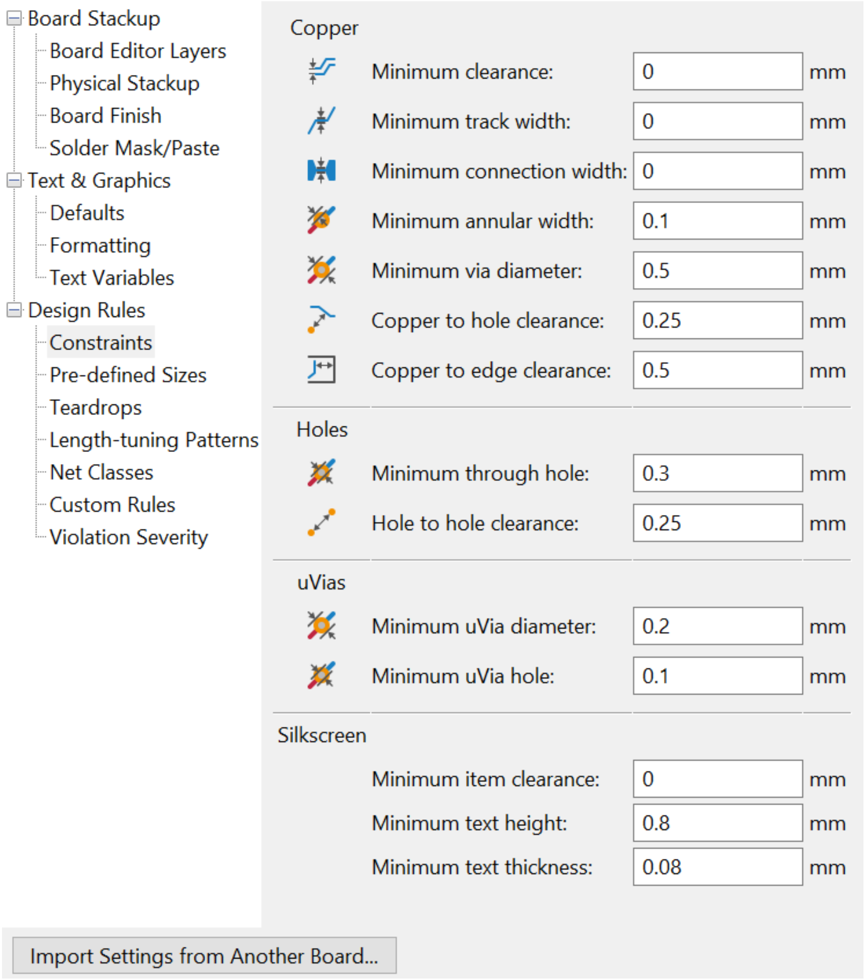


Galvanische Innenlage /inner layers (galvanic)



ELEKONTA MAREK[52]

Manufacturer Constraints



Drills (Drill)

PCB-thickness in mm	Calc. Finished holesize Aspect Ratio 1/8 * in mm	Min. Finished holesize in mm
1,0	0,13	0,15
1,2	0,15	0,15
1,4	0,18	0,2
1,6	0,2	0,2
1,8	0,23	0,25
2,0	0,25	0,25
…	…	…
4,8	0,6	0,6

* Smaller values are possible.

ELEKONTA MAREK[55]

Board Setup Menu → Solder Mask/Paste

KiCad

Board Setup

- Board Stackup
  - Board Editor Layers
  - Physical Stackup
  - Board Finish
  - Solder Mask/Paste
- Text & Graphics
  - Defaults
  - Formatting
  - Text Variables
- Design Rules
  - Constraints
  - Pre-defined Sizes
  - Teardrops
  - Length-tuning Patterns
  - Net Classes
  - Custom Rules
  - Violation Severity

Use your board manufacturer's recommendations for solder m
If none are provided, setting the values to zero is suggested.

Solder mask expansion: 0 mm

Solder mask minimum web width: 0 mm

Solder mask to copper clearance: 0 mm

☐ Allow bridged solder mask apertures between pads within footprint

☑ Tent vias

Solder paste absolute clearance: -0 mm

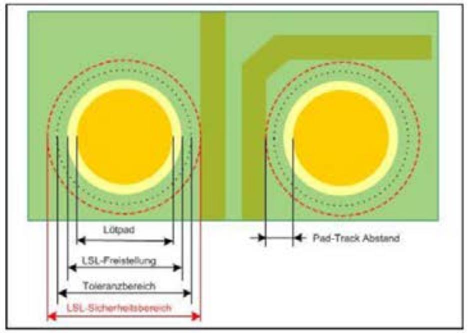
Solder paste relative clearance: -0 %

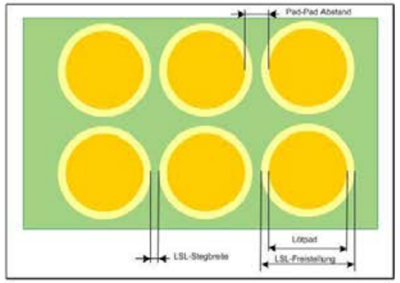
Note: Solder paste clearances (absolute and relative) are added to determine the

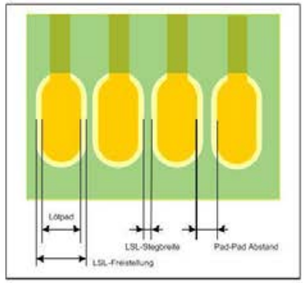
Manufacturer Constraints

Solder mask (Mask), Outer layer(Outer)

	Soldermask Clearence (50µm)	Soldermask Clearence (30µm)
Solderpad	X_P	X_P
Soldermask-Pad (negativ)	X_P +100µm	X_P +60µm
Distance Pad-Track	120µm	100µm
Min. Soldermask web	100µm	100µm
Distance Pad-Pad	200µm	160µm



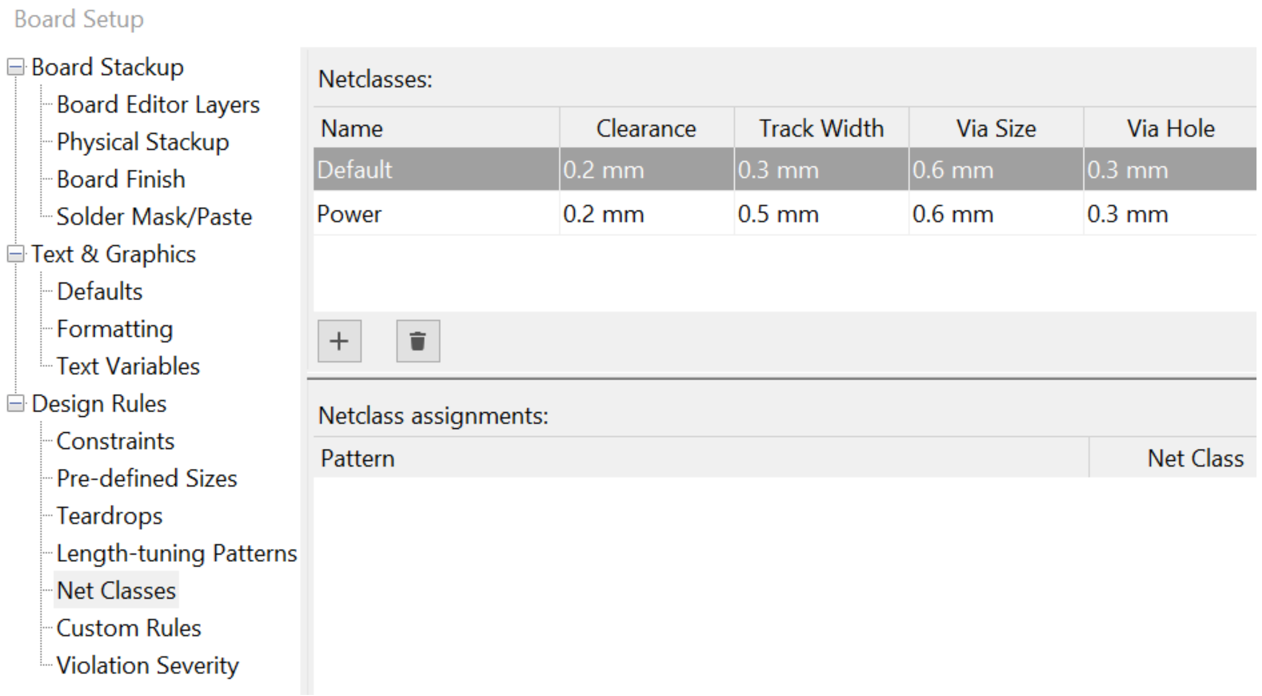




ELEKONTA MAREK

Based on these constraints, add two track widths

- 0.3mm
- 0.5mm



Agenda

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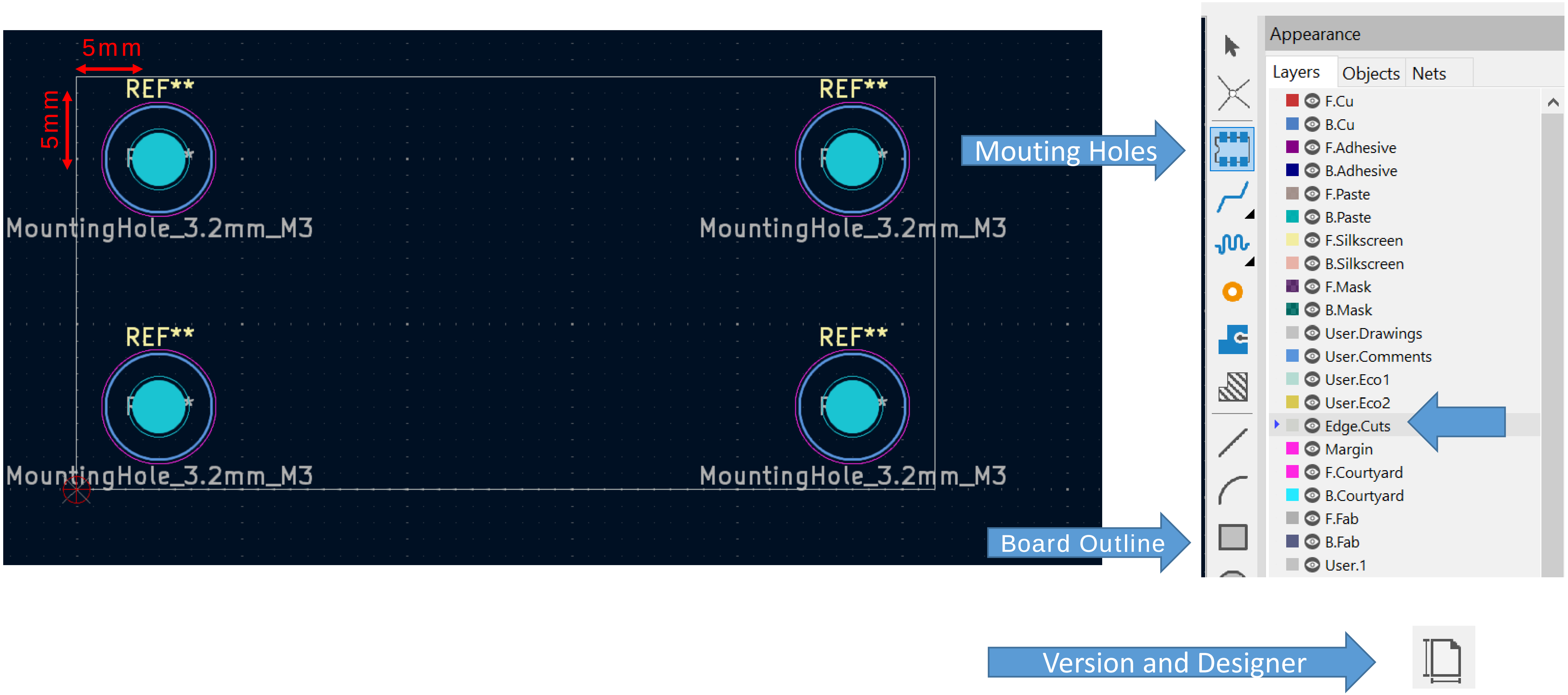
- Updating your Schematic
- Layout Constraints
- Mechanical Aspects
- Routing
- Prepare for Production
- Exercise
- Engineering Best Practice


5mm
5mm
REF**
REF**
REF**
REF**
MountingHole_3.2mm_M3
MountingHole_3.2mm_M3
MountingHole_3.2mm_M3
MountingHole_3.2mm_M3
Mouting Holes
Board Outline
Appearance
Layers
Objects
Nets
F.Cu
B.Cu
F.Adhesive
B.Adhesive
F.Paste
B.Paste
F.Silkscreen
B.Silkscreen
F.Mask
B.Mask
User.Drawings
User.Comments
User.Eco1
User.Eco2
Edge.Cuts
Margin
F.Courtyard
B.Courtyard
F.Fab
B.Fab
User.1
Version and Designer

Agenda

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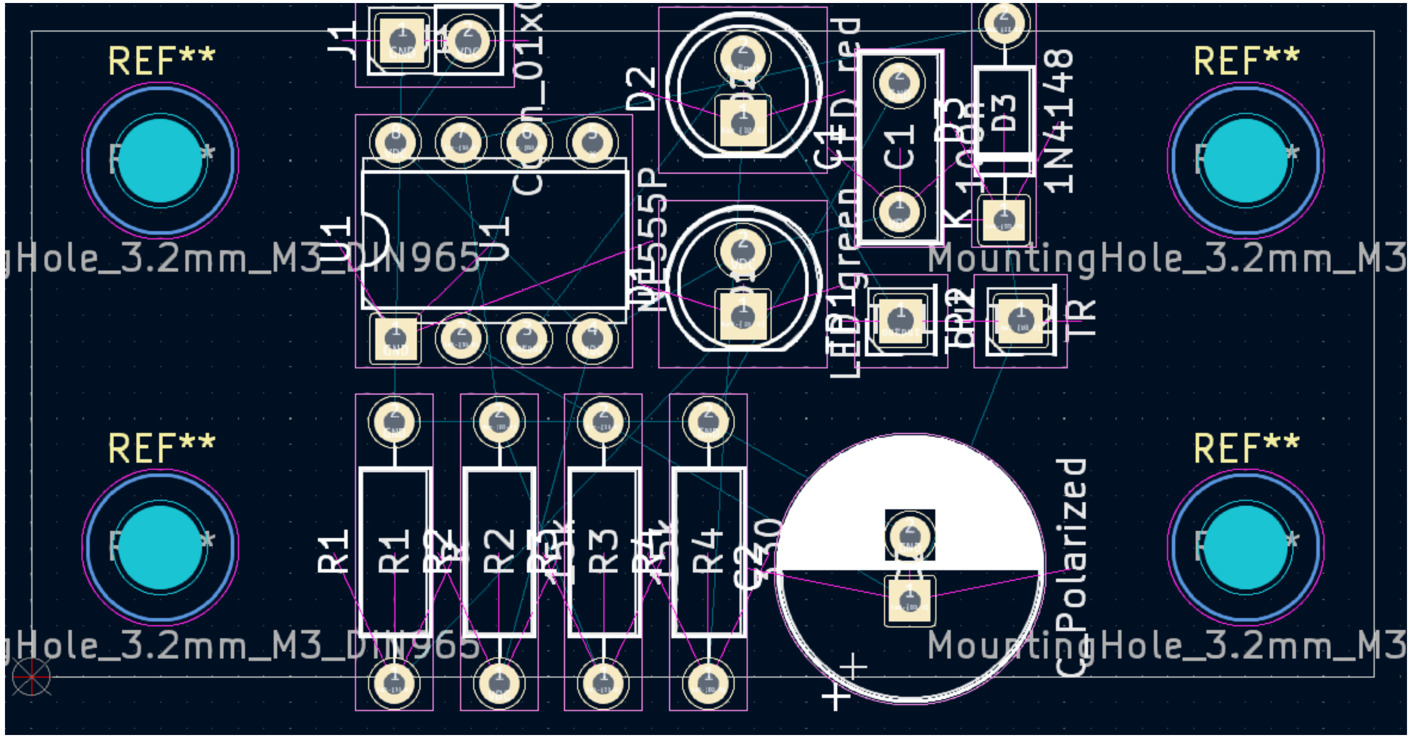
Routing: Import Netlist



Layout after importing the netlist

Rember the Layouting Constraints

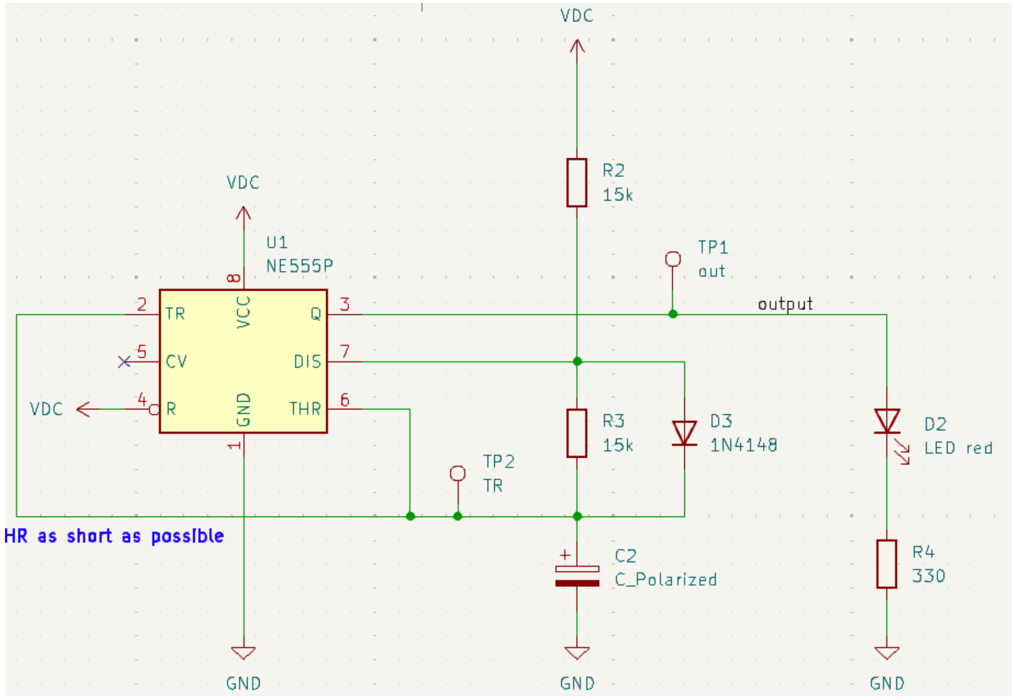
Layer:	2 copper layers
Top Layer:	signal routing
Bottom Layer:	GND – plane

Aim: resolve all crossings of the fly wires

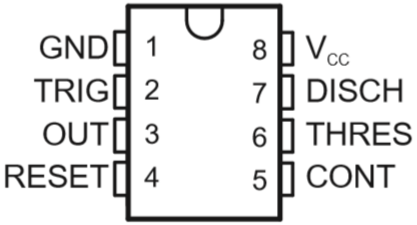
Strategy

- Keep in mind: the bottom Plane is the ground plane
- Changing the order of components to resolve crossings

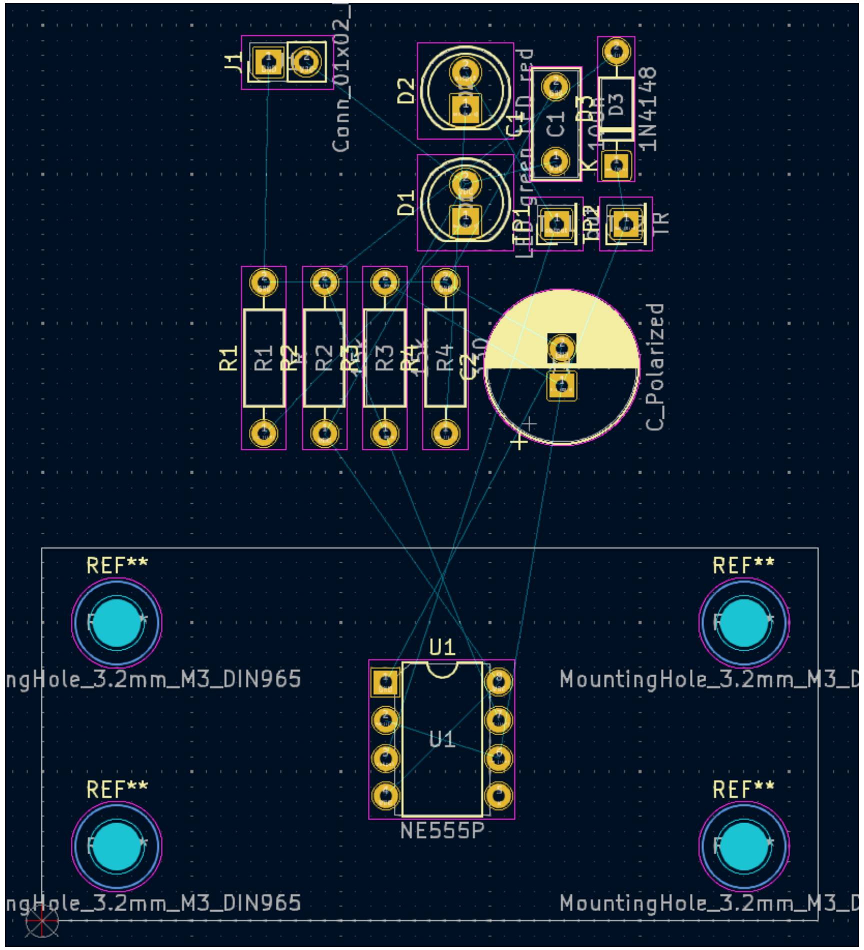
Routing: Place Components



Most relevant part of your circuit

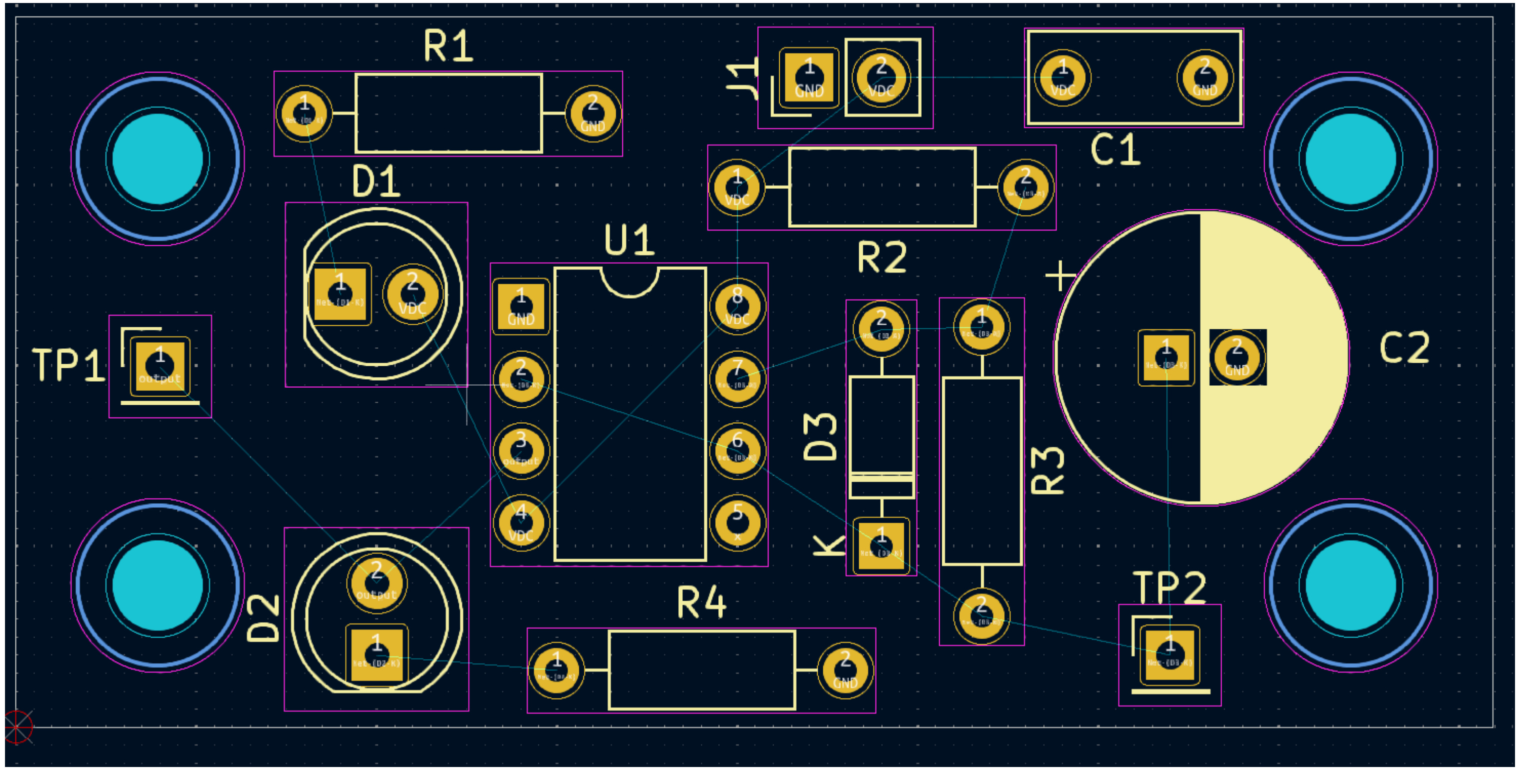


Footprint of your core unit



Starting Point for Placing of the components

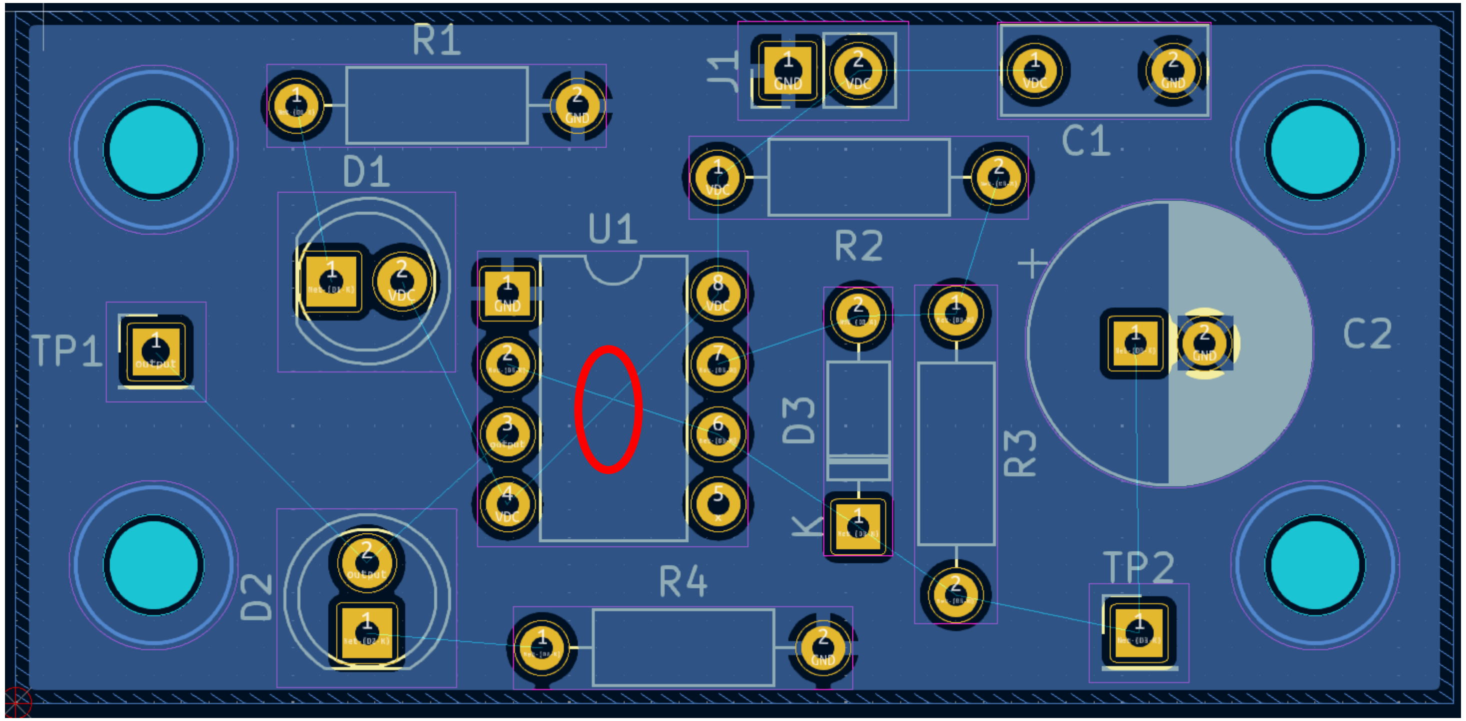
Routing: Poor the Ground Plane



Layout after all components have been placed

After the Component Placing
- o Assembling ?
- o Access to the test points ?
- o Make sure all texts are readable in landscape and 90 degrees turned.

Next Step
- o Poor the ground plane



Layout after filling the bottom layer and connecting it to the ground net

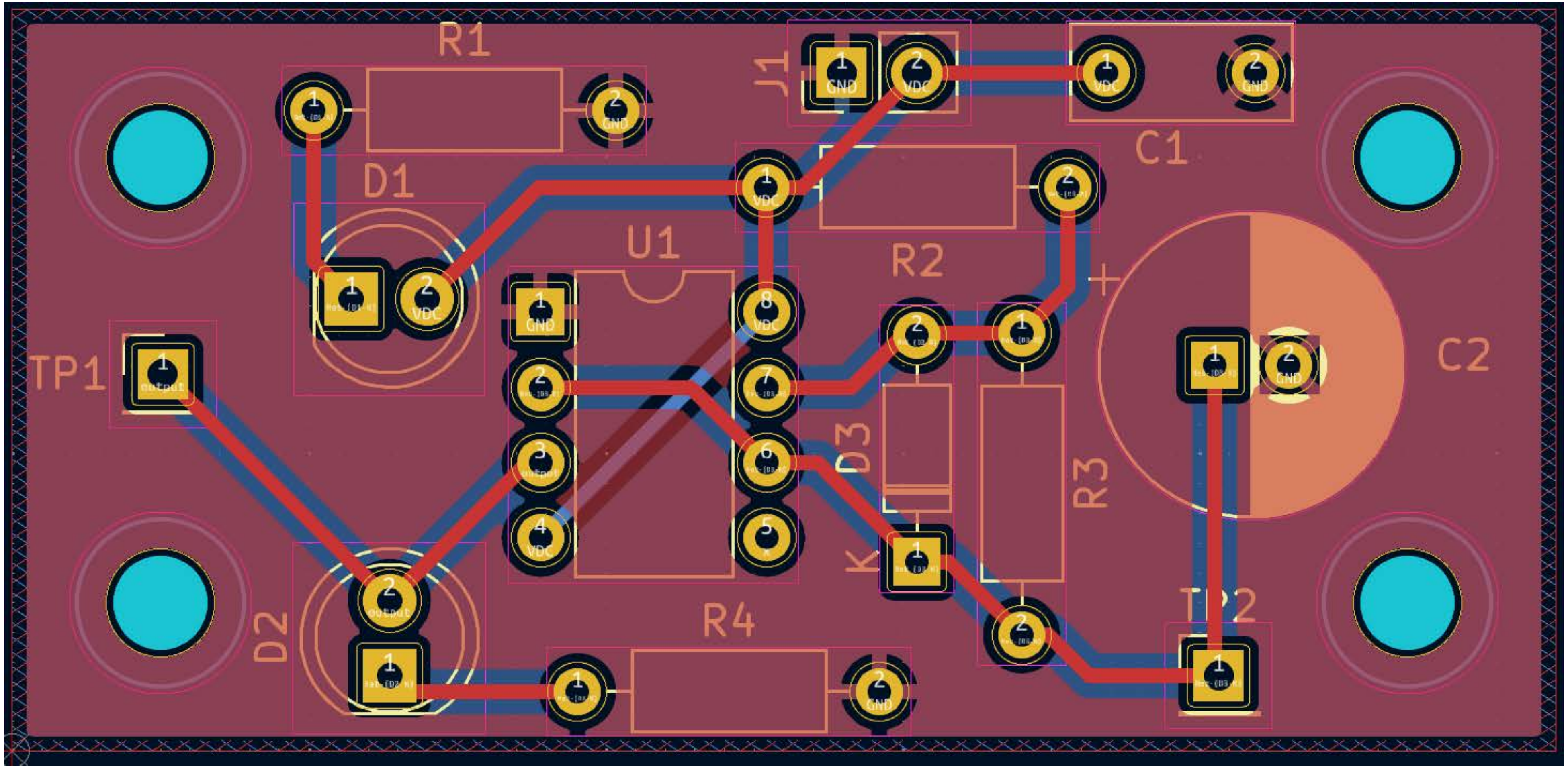
Strategy

- Left Crossings might be resolved by
  1. Routing between pins
  2. Routing under components
  3. If possible, you can route the power supply also on the bottom layer

Next Steps

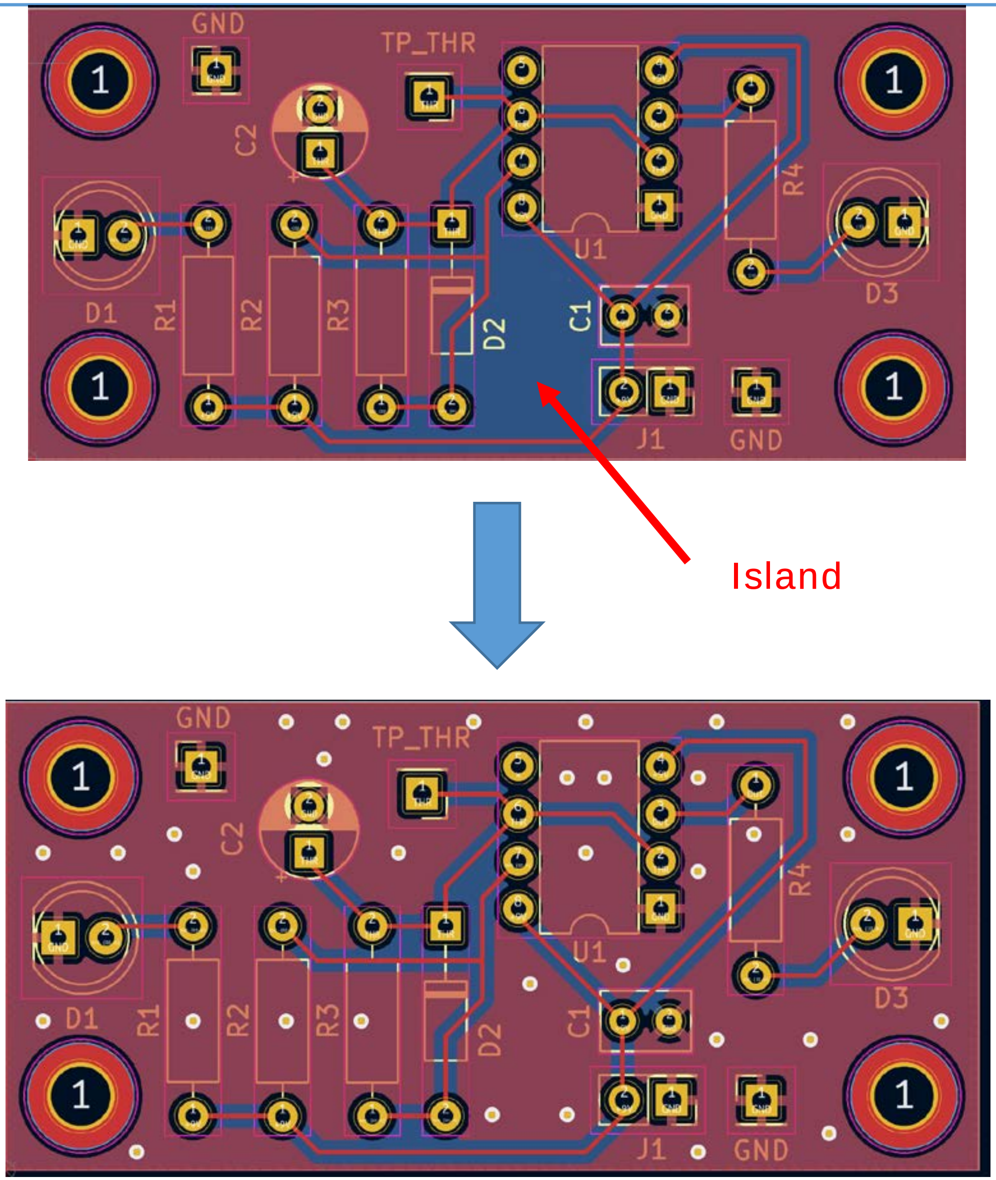
1. Signal routing
2. Pooring Ground plane on the front side

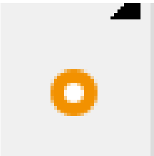
Routing: Finished



Add Freestanding – Vias | Pooring with Islands

hochschule darmstadt





Why should you add freestanding vias?

- You achieve a good ground connection (low resistive) between the different gorund planes.

How to taggle Islands?

- Add free-standing vias
- Fill again the ground plane

Next Step

- Poor the ground plane

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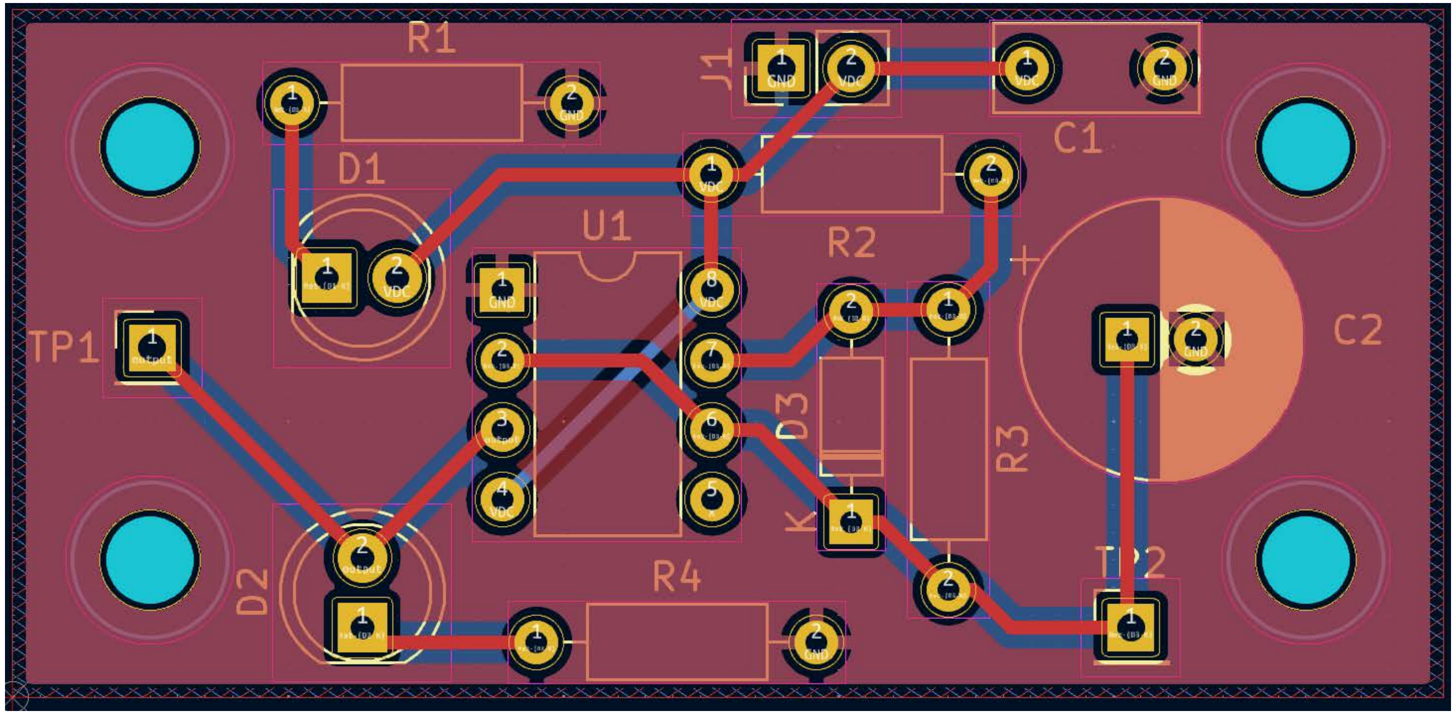
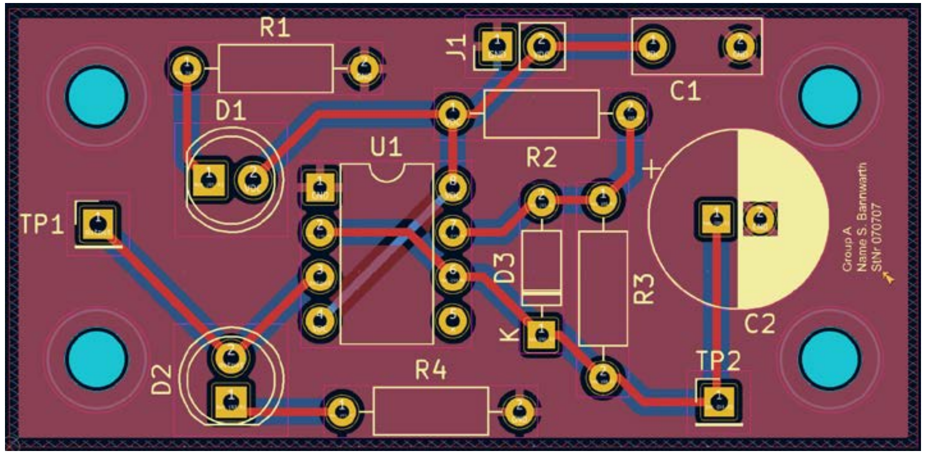


Next Steps

- Add Project Information on the Silk Layer
- Perform the DRC



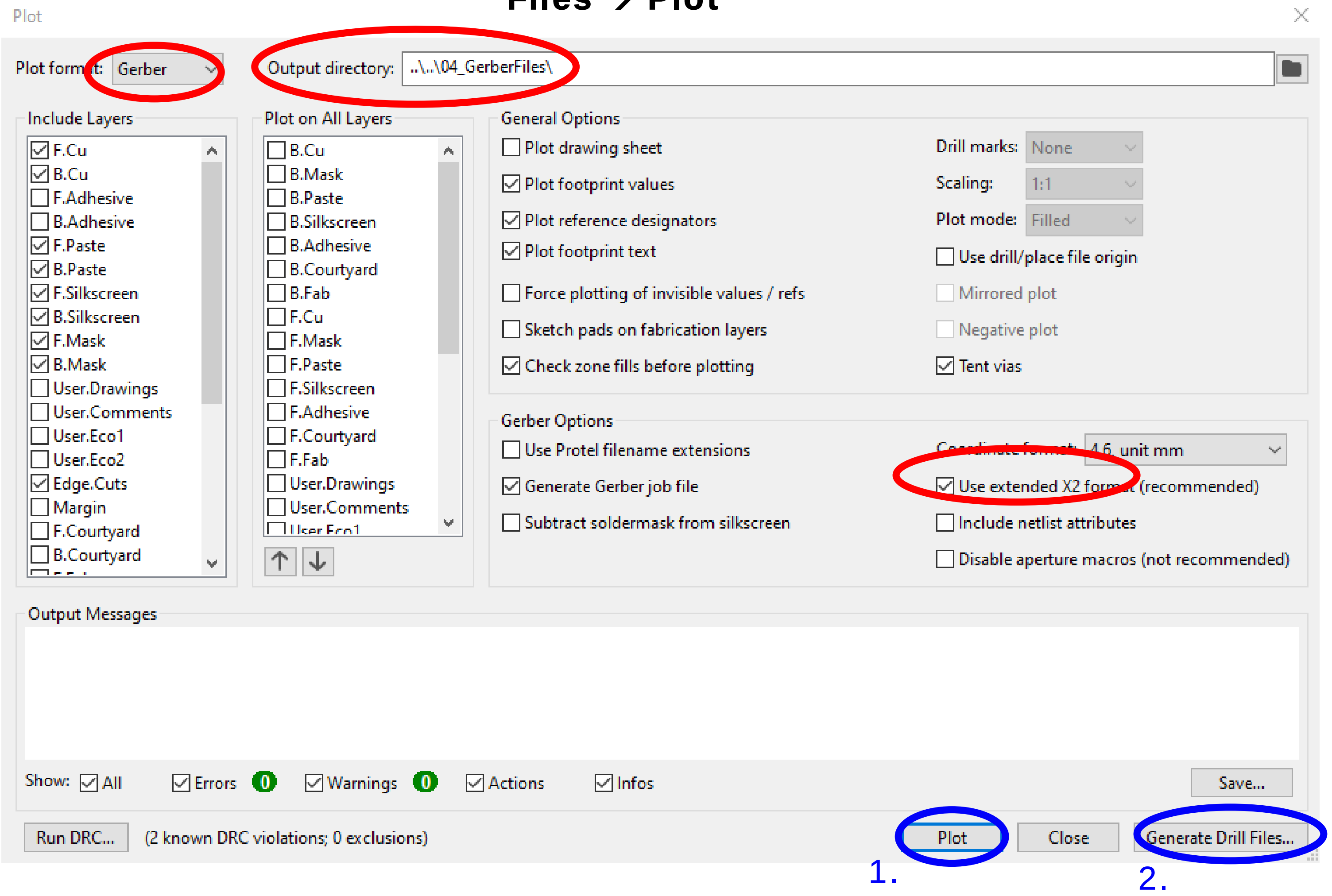
Board Ready for Gerber File Export

Students from Darmstadt

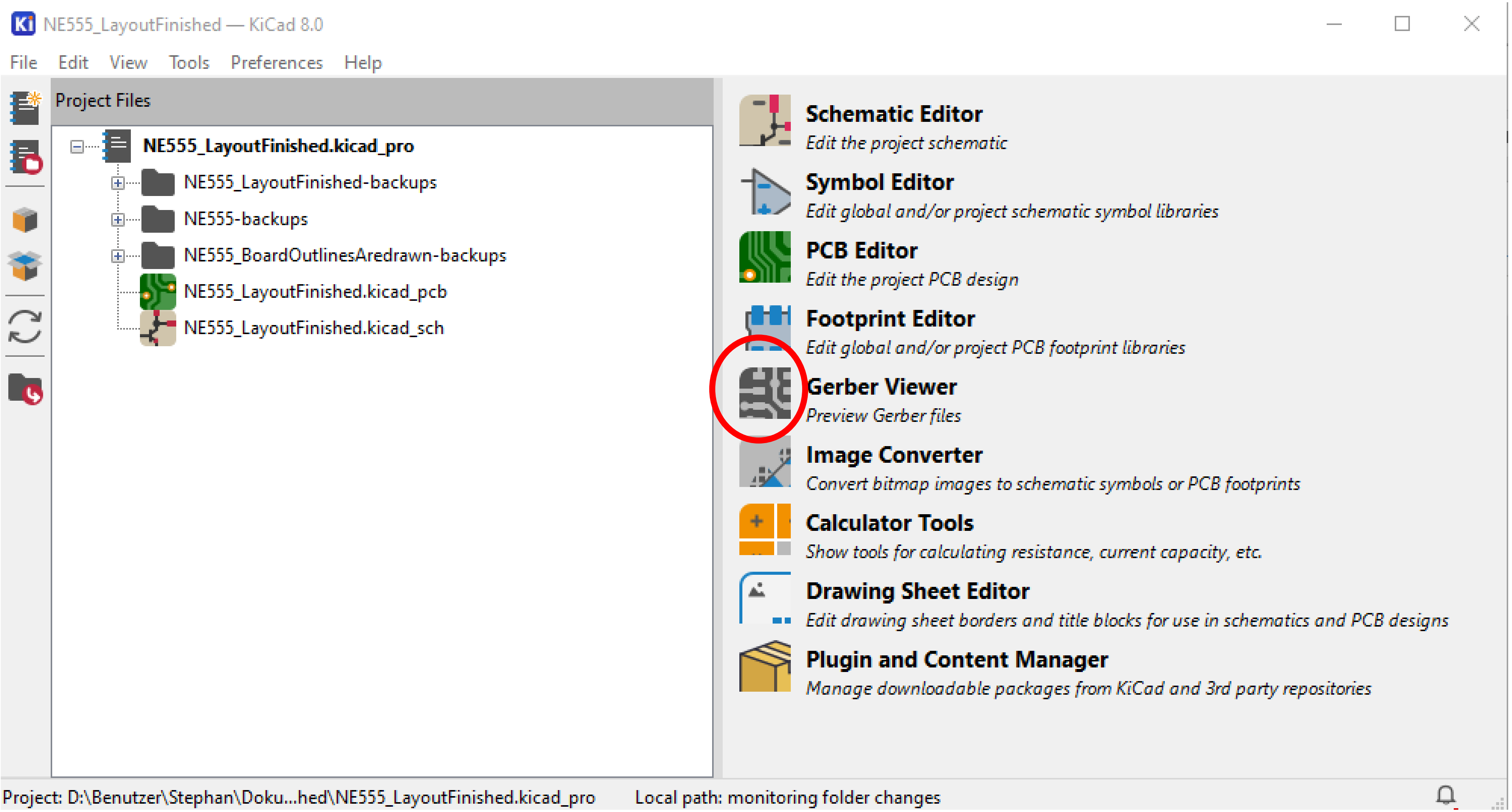
→ You will get information in our courses, whiche data you have to export

→ Check the moodle course

Files → Plot



Prepare For Production: Gerber Viewer



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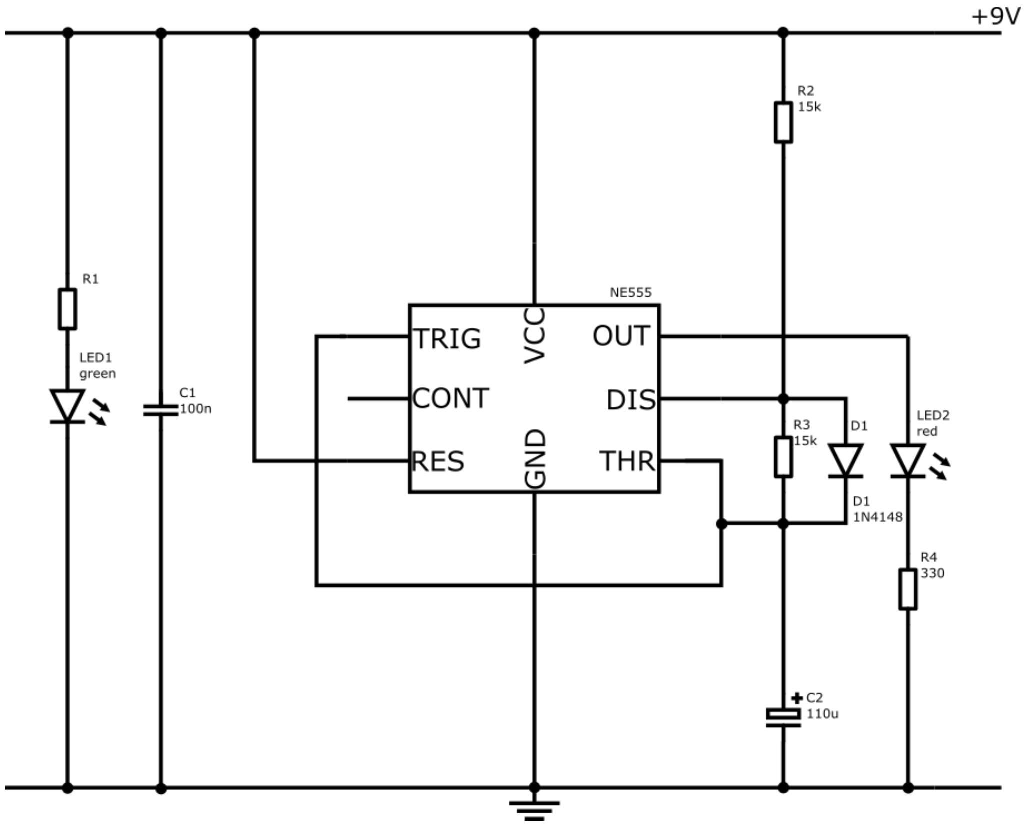


PCB Constraints

Board Size:	52mm x 25mm
4x Mounting Holes:	M3
Layer:	2 copper layers
Top Layer:	signal routing
Bottom Layer:	GND-plane and VCC

Update your Schematic

1. Insert Testpoints
2. New Version of the worksheets

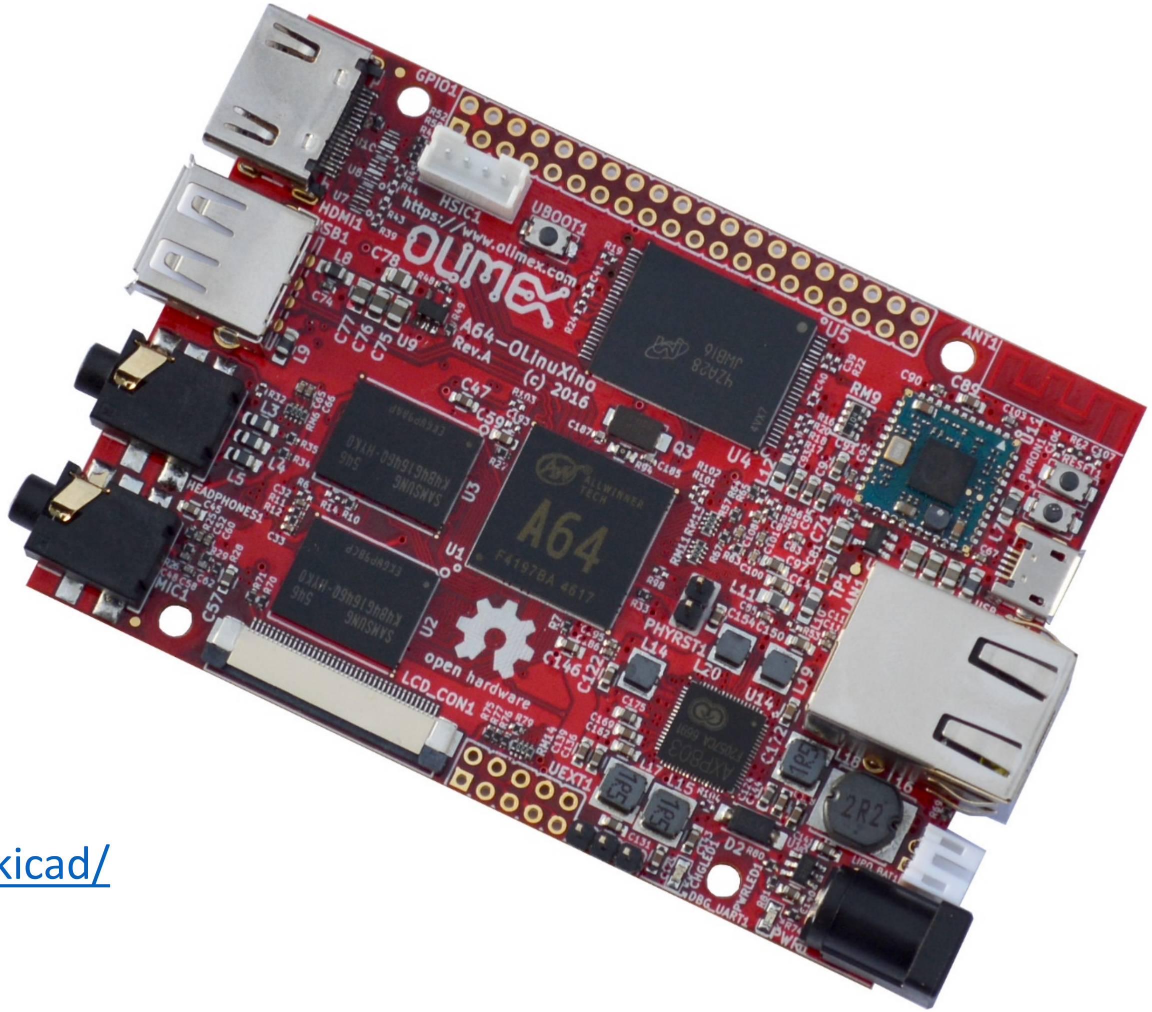
Agenda

- Layout Task and its Working Steps
- Updating your Schematic
- Layout Constraints
- Mechanical Aspects
- Routing
- Prepare for Production
- Exercise
- Engineering Best Practice

Engineering Best Practice

1. Ask yourself and your team questions
   a) How to test?
   b) How to mount?
   c) How to assemble?
2. Update your schematic
3. Before you start to layout
   a) Print out the main circuit
   b) Print out the main component's footprint
   → Make a rough planning where you place the different sub circuits
4. Enter the mechanical Aspects first
   a) Edge Cut
   b) Mounting holes
   c) Place the connectors
5. Put the Project Information on your silk screen (Project Name, Version, Author, Date)
6. Run DRC
7. Export Gerber and Drill files
8. Always Check with the Gerber Viewer

A64-OLinuXino
Olimex
A64-OLinuXino is a single-board computer running Linux & Android. The design based on a 64-bit ARM CPU and includes 1 or 2 GB DDR3 RAM, 4 GB flash memory, microSD card socket, WiFi & BLE4.0, Ethernet, HDMI & audio output.



https://kicad-pcb.org/made-with-kicad/

Thank you for Watching
&
Have Fun with your PCB – Design